TEXORA · LECTURE SHEET

# Production Calculation: IE, Capacity and Efficiency

*Industrial engineering arithmetic: how an SMV is built from a time study, and how SMV, manpower and efficiency determine a line's target, capacity and pitch time.*

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 7 · SPECIALIZATION — APPAREL MANUFACTURING

## 1. The standard minute value (SMV)

All production calculation runs on one unit: the **standard minute value** of an operation or garment — the time a qualified operator, working at standard pace with standard method, needs for the work, including allowances for rest and unavoidable delay. An SMV is built, not guessed:

Observed time → Basic time = observed × (rating ÷ 100) → SMV = basic time × (1 + allowance %)

**WORKED EXAMPLE — BUILDING ONE OPERATION'S SMV**

A time study of a collar-attach operation records an observed time of **0.42 min**. The observer rates the operator at **110%** of standard pace. Basic time = 0.42 × 1.10 = 0.462 min. Relaxation and contingency allowances total **15%**: SMV = 0.462 × 1.15 = 0.5313 ≈ **0.53 min**. The garment SMV is the sum of its operation SMVs — for the course shirt, the operation breakdown sums to **SMV 20 min**, the value used throughout this sheet and Topic 8.

## 2. Capacity, target and pitch

Line capacity (pcs/day) = (operators × minutes worked) ÷ SMV · Target = capacity × efficiency

Pitch time = SMV ÷ operators (minutes per piece at the line's station rhythm)

**WORKED EXAMPLE — THE COURSE SHIRT LINE**

Line: **30 operators**, 8-hour shift = 480 min, shirt SMV **20 min**.

Capacity at 100% = (30 × 480) ÷ 20 = **720 pcs/day**; hourly target at 100% = 720 ÷ 8 = **90 pcs/h**.

At the factory's working efficiency of 65%: target = 720 × 0.65 = **468 pcs/day**.

> Pitch time = 20 ÷ 30 = 0.667 min = **40 s** — a finished shirt must leave the line every 40 seconds at full pace; every workstation is balanced against this rhythm in Topic 8.
> Check by earned minutes: 468 pcs × 20 SMV = 9,360 standard minutes earned against 30 × 480 = 14,400 minutes attended: 9,360 ÷ 14,400 = 65% ✓.

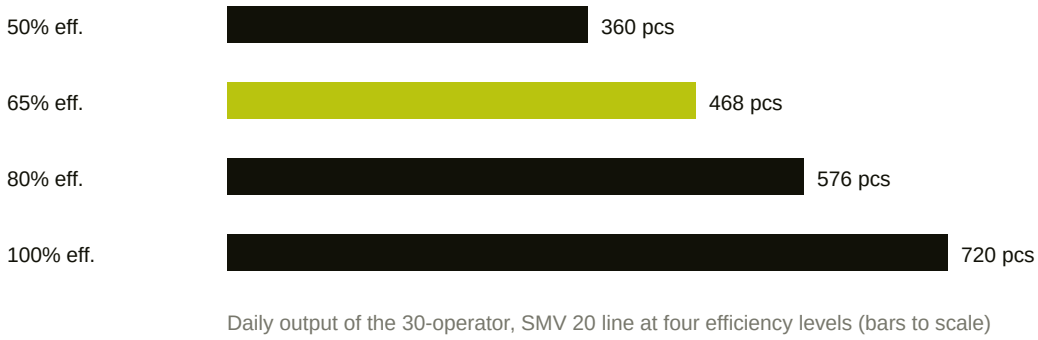



Figure 1 — Efficiency is a multiplier on capacity, never an addition: the same line, machines and SMV produce 360 or 720 shirts a day depending on it.

## 3. Operator efficiency

The same arithmetic grades an individual. An operator sewing an operation of SMV 5 min produces 60 pieces in a 480-minute shift: earned minutes = 60 × 5 = 300; efficiency = 300 ÷ 480 × 100 = **62.5%**. Individual efficiency feeds line balancing (Topic 8): operators are assigned to operations so that each station's load matches its operator's demonstrated pace, and low-efficiency operations are the first candidates for the work aids and automation of AM-3207 Topic 1 and TE-3107.

## 4. What efficiency is made of

Efficiency losses are classified so they can be attacked: **machine and waiting losses** (breakdowns, thread breaks, bundle waiting), **method losses** (poor workplace layout, unnecessary handling — reduced by work study), **learning losses** (a new style's output climbs a learning curve over its first days, which is why short orders quote lower efficiency), and **quality losses** (alteration and rework consume minutes that produce no new garments — the link to Topic 9). A quoted target that ignores the learning curve on day one of a new style is not a plan; it is a hope.

## Glossary

| TERM | MEANING |
|---|---|
| SMV / SAM | Standard minute (allowed) value: standard time for an operation or garment, allowances included. |

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|---|---|
| Rating | The observer's assessment of an operator's pace as a percentage of standard pace, used to normalise observed time. |
| Efficiency | Standard minutes earned ÷ minutes attended × 100, for an operator, line or factory. |
| Pitch time | SMV ÷ operators: the interval at which finished pieces must leave the line. |
| Capacity | Output at 100% efficiency: (operators × minutes) ÷ SMV. |

## Key points and summary

- SMV = observed time × rating × (1 + allowances); garment SMV is the sum of operation SMVs (course shirt: 20 min).
- Capacity = operators × minutes ÷ SMV; target = capacity × efficiency; pitch = SMV ÷ operators.
- Course line: 720 pcs/day at 100%, 468 at 65%, pitch 40 s.
- Efficiency losses are classified — machine, method, learning, quality — so each can be assigned a remedy.

## Practice

Answer in your own words; brief model answers follow.

1. An operation is observed at 0.36 min with a rating of 90% and allowances of 15%. Build its SMV.
2. A 25-operator line makes a garment of SMV 16 min in 480-minute shifts. Calculate capacity at 100% and the target at 70% efficiency, and the pitch time.
3. An operator produces 95 pieces of an SMV 3.2 min operation in 480 minutes. Calculate her efficiency.
4. Why does a factory quote a lower efficiency for the first two days of a new style even with experienced operators?

### Answer key

1. Basic = 0.36 × 0.90 = 0.324 min; SMV = 0.324 × 1.15 = 0.3726 ≈ 0.37 min.
2. Capacity = 25 × 480 ÷ 16 = 750 pcs/day; target at 70% = 525 pcs/day; pitch = 16 ÷ 25 = 0.64 min = 38.4 s.
3. Earned = 95 × 3.2 = 304 min; efficiency = 304 ÷ 480 × 100 = 63.3%.
4. Learning loss: operators must build new movement patterns, machine settings and quality judgements for the unfamiliar style, so earned minutes lag attended minutes until the curve flattens — experienced operators climb faster, not instantly.